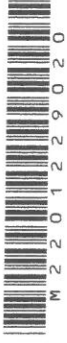


CARIBBEAN EXAMINATIONS COUNCIL

CARIBBEAN SECONDARY EDUCATION CERTIFICATE®
EXAMINATION

24 MAY 2022 (a.m.)



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FILL IN ALL THE INFORMATION REQUESTED CLEARLY IN CAPITAL LETTERS.

TEST CODE

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SUBJECT INFORMATION TECHNOLOGY – Paper 02

PROFICIENCY GENERAL

REGISTRATION NUMBER

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SCHOOL/CENTRE NUMBER

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

NAME OF SCHOOL/CENTRE

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CANDIDATE'S FULL NAME (FIRST, MIDDLE, LAST)

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DATE OF BIRTH

D	D	M	M	Y	Y	Y	Y
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MAY/JUNE 2022

C A R I B B E A N E X A M I N A T I O N S C O U N C I L

**C A R I B B E A N S E C O N D A R Y E D U C A T I O N C E R T I F I C A T E[®]
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Paper 02 – General Proficiency

2 hours

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This paper consists of FOUR questions. Answer ALL questions.
2. Write your answers in the spaces provided in this booklet.
3. Do NOT write in the margins.
4. If you need to rewrite any answer and there is not enough space to do so on the original page, you must use the extra lined page(s) provided at the back of this booklet. **Remember to draw a line through your original answer.**
5. If you use the extra page(s) you **MUST** write the question number clearly in the box provided at the top of the extra page(s) and, where relevant, include the question part beside the answer.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

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01229020/MJ/CSEC 2022



0 1 2 2 9 0 2 0 0 3



Answer ALL questions.

1. (a) Direct data entry devices can transfer data directly from a document into a computer system.

- (i) Circle TWO direct data entry devices from the following list of input devices.

Barcode reader	Light pen	Optical mark reader
Joystick	Touchscreen	(2 marks)

- (ii) Name an application in which any ONE of the devices circled in (a) (i) is used and describe the type of data that is captured with that device.

.....
.....
(2 marks)

- (b) Turnaround documents are popular with automated methods of data entry.

- (i) Describe what is meant by a turnaround document.

.....
.....
.....
(2 marks)

- (ii) Explain how data is captured from a turnaround document.

.....
.....
(2 marks)



- (c) Determine the number of 5 MB documents that could fit on a 16 GB flash drive.
(Show all working.)

.....

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.....

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(2 marks)

- (d) Consider the following paragraph.

A printer is an output device on a computer system used to print hard copy output. Printers fall into two categories: impact and non-impact.

Non-impact printers spray ink or toner powder and the characters are fixed onto the paper by heating. Examples of non-impact printers are inkjet printers, laser printers and thermal printers.

Impact printers strike the paper when printing. The most popular impact printer is a dot-matrix printer. Dot-matrix printers print multiple copies using carbon paper.

- (i) State the steps (in the correct order) required to find the number of words in the paragraph above, using a word processing program.

.....

.....

.....

(2 marks)

- (ii) List the steps (in the correct order) required to conduct a spell-check on the paragraph above, using a word processing program.

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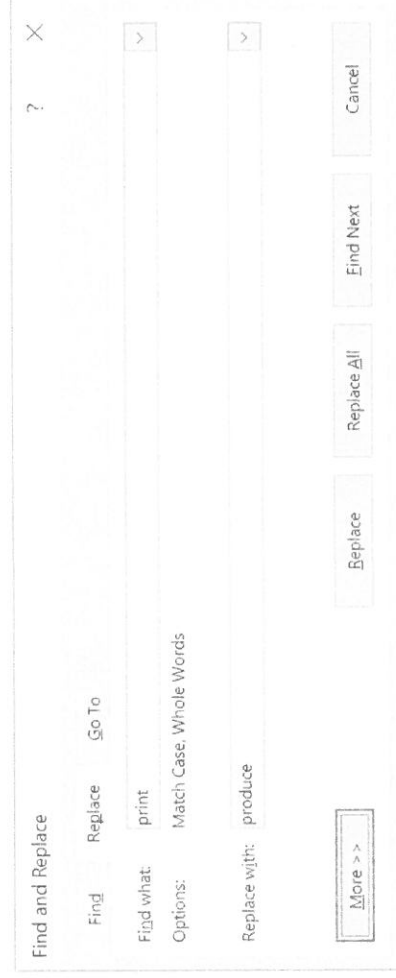
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(3 marks)

GO ON TO THE NEXT PAGE



- (e) The following diagram illustrates the dialogue box obtained when using the Find and Replace editing feature in a word processing program, to replace all occurrences of the word “print” with the word “produce” in the paragraph in (d) on page 5.



The diagram shows a 'Find and Replace' dialog box. It has a title bar 'Find and Replace' with a question mark and a close button. The 'Find' tab is selected. The 'Find what:' field contains 'print'. The 'Replace with:' field contains 'produce'. The 'Options:' section has 'Match Case, Whole Words' selected. The 'Go To' button is visible. At the bottom, there are buttons for 'More >>', 'Replace', 'Replace All', 'Find Next', and 'Cancel'.

- (i) Briefly explain why it would be safer to use the Replace option rather than the Replace All option in the dialogue box above.

.....

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.....

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(2 marks)

- (ii) Outline the steps needed to replace all occurrences of the word “print” with the word “produce” using the Replace All option in the dialogue box above.

.....

.....

.....

.....

(3 marks)

Total 20 marks



2. Sharon recently had a home network built with both wired and wireless capability to facilitate her new home-based business initiative.

- (a) State the location on the network where EACH of the following hardware devices could be found AND describe the function of EACH device.

Network interface card

Location:

Function:

.....

Wireless network adaptor

Location:

Function:

.....

Modem

Location:

Function:

.....

.....

Router

Location:

Function:

.....

.....

(8 marks)



- (b) The wireless devices were accessing the Internet but Sharon's desktop computer was unable to access the Internet. Suggest TWO instructions that could be given to Sharon to help her to troubleshoot the problem.

.....

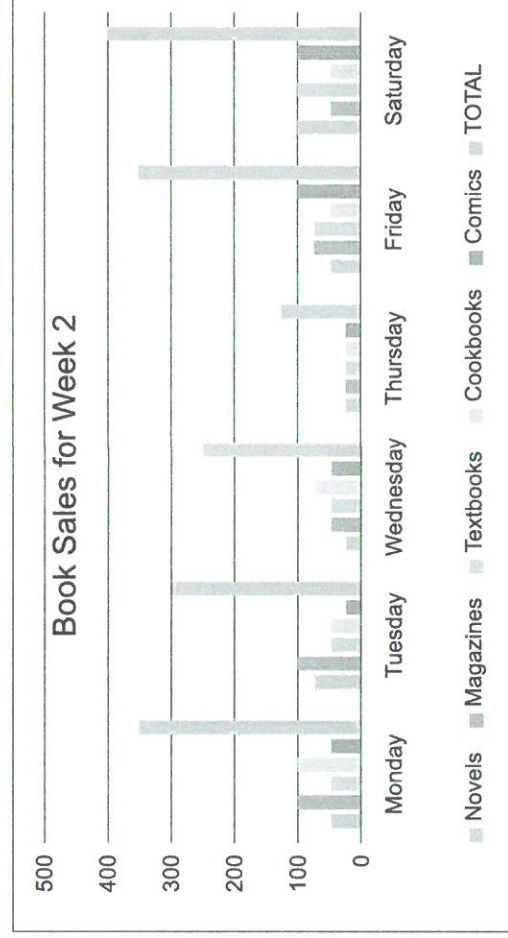
.....

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(2 marks)

- (c) Examine the following chart created from data in a spreadsheet.



- (i) List the steps (in the correct order) required to create the chart from data in a spreadsheet.

.....

.....

.....

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.....

.....

(4 marks)



- (ii) State ONE other type of chart which is appropriate to represent the data in the spreadsheet and state how it differs from the one on page 8.

.....

.....

.....

.....

(2 marks)

- (iii) Name ONE type of chart **other** than those mentioned in (c) (i) and (ii) and describe when it is BEST used.

.....

.....

.....

(2 marks)



- (iv) Sharon used the following spreadsheet to create the chart displayed on page 8.

	A	B	C	D	E	F	G	H
1	Books Sold for Week 2							
2	Type of Book	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Total
3	Novels	50	75	25	25	50	100	325
4	Magazines	100	100	50	25	75	50	375
5	Textbooks	50	50	50	25	75	100	350
6	Cookbooks	100	50	75	25	50	50	350
7	Comics	50	25	50	25	100	100	350
8	TOTAL		300	250	125	350	400	1775

Write the formula/function to be inserted in Cell B8, to calculate the total number of books sold on Monday.

.....

.....

.....

.....

(2 marks)

Total 20 marks



3. (a) Complete the table below by inserting the MOST appropriate web technology concepts from the following list.

Web browser Hypertext markup language Hypertext transfer protocol
Hyperlink Web server File transfer protocol Uniform resource locator

Web Technology Concept	Description
	Connects webpages to other webpages
	Stores, processes and delivers webpages to client devices on request
	Facilitates communication with client devices for the delivery of webpages
	Software application that allows client devices to access resources and websites from the Internet
	Text-based application used to create web pages for display on the client devices

(5 marks)



Consider the following database with two database tables named STUDENTS and SPORTS.

STUDENTS

ID	FirstName	LastName	Birthdate	Gender	Married	Phone	SportNo
1	Roy	CLARKE	1/2/1997	M	Yes	688-4444	010
2	Jamila	ALI	3/22/1982	F	No	720-9123	020
3	Tyson	LENNY	3/23/1996	M	Yes	630-1234	040
4	Ikelia	DRAKES	7/4/1978	F	No	456-7890	050
5	Suresh	BAPTISTE	5/23/1999	M	No	456-6676	030
6	Janet	BROWN	6/18/2001	F	Yes	620-0190	050
7	Smith	JONES	12/20/2002	M	Yes	640-2823	020
8	Michael	JONES	6/18/1999	M	Yes	640-4929	080
10	Daniella	SINGH	6/18/2008	F	No	640-9989	090
11	Bridget	NICHOLAS	6/18/2008	F	No	637-4442	100
14	Brian	JOSEPH	3/3/2001	M	Yes	638-2223	020

SPORTS

SportNo	SportName
010	Cricket
020	Football
030	Basketball
040	Lawn tennis
050	Badminton



(b) The database tables STUDENTS and SPORTS can be joined in a one-to-many relationship to ensure that data in the two tables remain as accurate as possible.

(i) State the feature that is used to establish and maintain the accuracy of data between the two database tables.

.....

.....

(1 mark)

(ii) State why it is necessary to maintain the accuracy of data between the two database tables.

.....

.....

(1 mark)

(iii) Outline how the one-to-many relationship is formed between the two database tables.

.....

.....

.....

(2 marks)

(c) Create a query by completing the following query design view window to extract

- the fields **ID**, **FirstName**, **LastName** and **Phone** from the STUDENTS table
- the field **SportName** from the SPORTS table, for all married students who play football.

Field	ID	FirstName	LastName	Married	Phone	SportNo	SportName
Table	STUDENT	STUDENT	STUDENT	STUDENT	STUDENT	SPORTS	SPORTS
Sort							
Show	☐	☐	☐	☐	☐	☐	☐
Criteria							
OR							

(4 marks)



- (d) State the record IDs of the record(s) that would be extracted when the query designed in (c) is executed.

.....

.....

(2 marks)

- (e) (i) There are four steps in implementing a program. List THREE of these steps.

.....

.....

.....

(3 marks)

- (ii) Describe what is involved in any TWO of the steps listed in (e) (i).

.....

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.....

.....

(2 marks)



L

- # START

Read Team_A, goals_A

Read Team_B, goals_B

Total 25 marks

4. A credit union gives loans of \$10 000 at a fixed monthly interest rate of one percent on the principal amount. The monthly payment amount is a fixed \$1000. The amount paid to the principal each month is the monthly payment amount minus the interest paid for the month.

The pseudocode below will print a payment schedule showing the principal, the payment made to the principal and the interest charged for each month once the principal is greater than zero. The payment schedule will also print the total amount paid to the principal, the total amount of interest paid and the number of payments made.

```
START
Line 1.      Set Principal ← 10,000
Line 2.      Set MonthlyPayment ← 1,000
Line 3.      Set MonthlyInterest ← 0.01

Line 4.      Write "Principal", "Payment", "Interest"

Line 5.      WHILE Principal > 0 DO
Line 6.        Set Interest ← Principal * MonthlyInterest
Line 7.        Set Payment ← MonthlyPayment - Interest
Line 8.        Set TotalInterest ← TotalInterest + Interest
Line 9.        Set TotalPayment ← TotalPayment + Payment

Line 10.     Write Principal, Payment, Interest

Line 11.     Set NewPrincipal ← Principal - Payment
Line 12.     Set Count ← Count + 1

Line 13.     END WHILE

Line 14.     Write TotalPayment, TotalInterest, Count
```

STOP

- (a) Identify TWO variables which when added will always be equal to the monthly repayment amount of \$1000.

.....
(2 marks)

- (b) (i) The pseudocode has a logic error. Identify the line with the error and make the necessary corrections.

.....
(2 marks)



- (ii) Identify the stage in program implementation where the type of error stated in (b) (i) will show up and state the consequences of the error.

.....

.....

(2 marks)

- (c) State the type of looping construct used in the pseudocode.

.....

(1 mark)

- (d) Suggest ONE reason why the looping construct stated in (c) was used in the pseudocode.

.....

.....

(2 marks)

- (e) Explain ONE other **named** looping construct that could be used in the program.

.....

.....

(2 marks)



- (f) Complete the following trace table by inserting the missing values for the variable in EACH row representing the first THREE times the pseudocode loops.

	Principal	Payment	Monthly Interest
Loop 1	\$ 10 000	_____	_____
Loop 2	_____	_____	_____
Loop 3	\$ 8 191	_____	\$ 81.91

(4 marks)

- (g) The advent of the COVID-19 pandemic has resulted in an increase in connectivity and online activities, for example, telecommuting, cyber-criminal activities and greater use of cloud-based applications.

- (i) Describe the concept of 'telecommuting' and state ONE reason why telecommuting is so impactful.

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(2 marks)

- (ii) Describe the term 'denial-of-service (DOS) attack', giving ONE example.

.....

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.....

.....

(3 marks)



(iii) State ONE similarity and ONE difference between Bluetooth and Wi-Fi technologies.

Similarity

Difference

(2 marks)

(iv) State TWO advantages and ONE disadvantage of buying software applications stored in the cloud.

Advantages

.....

.....

.....

Disadvantage

.....

(3 marks)

Total 25 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.

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EXTRA SPACE

If you use this extra page, you **MUST** write the question number clearly in the box provided.

Question No.

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